

1) Available Wi-Fi Networks

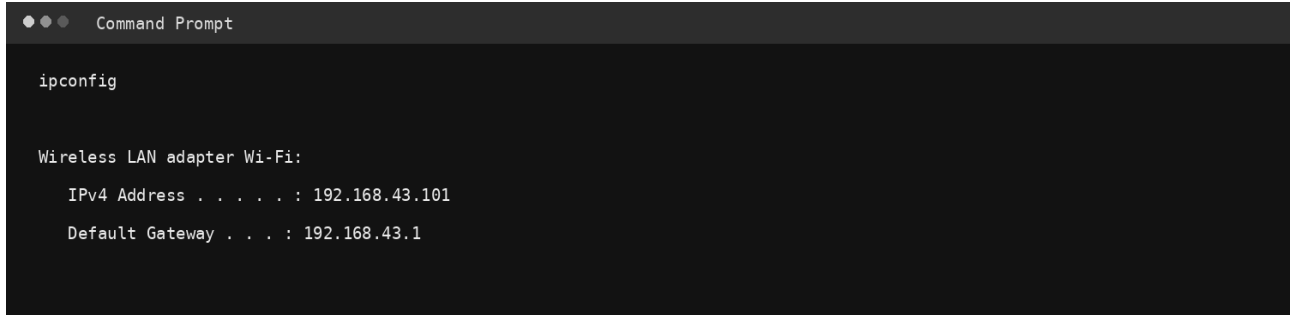
- Home_WiFi — WPA2/WPA3 — Strong signal.
- Airtel_XYZ — WPA2 — Medium signal.
- Cafe_Free_WiFi — Open / Unsecured — Weak signal.
- Neighbor_5G — WPA2 — Medium signal.

Available Wi-Fi Networks

SSID	Security	Signal
Home_WiFi	WPA2/WPA3	Strong
Airtel_XYZ	WPA2	Medium
Cafe_Free_WiFi	Open / Unsecured	Weak
Neighbor_5G	WPA2	Medium

2) Mobile Hotspot IP Address

- Hotspot IPv4 address received by the laptop: 192.168.43.101.
- Default gateway: 192.168.43.1.
- The laptop receives a private IP address from the phone's hotspot network.



```
Command Prompt

ipconfig

Wireless LAN adapter Wi-Fi:

    IPv4 Address . . . . . : 192.168.43.101
    Default Gateway . . . . : 192.168.43.1
```

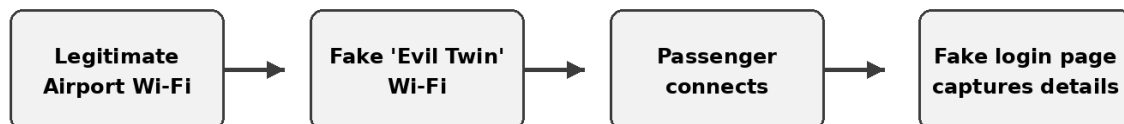
3) WPA2 and WPA3

- WPA2 is a widely used Wi-Fi security standard using strong encryption such as AES-CCMP.
- WPA3 is the newer standard and improves password-based authentication using SAE.
- Difference 1: WPA3 provides stronger protection against offline password-guessing attacks.
- Difference 2: WPA3 uses newer authentication mechanisms and improves protection for modern Wi-Fi networks.

Feature	WPA2	WPA3
Authentication	PSK / enterprise	SAE / enterprise
Password security	Good	Stronger against offline guessing
Encryption	AES-CCMP commonly used	Stronger modern protection

4) Public Wi-Fi Cyberattack — Evil Twin Case

- Case: In 2024, Australian police charged a man accused of creating fake free Wi-Fi networks at airports and on domestic flights.
- The fake networks copied legitimate Wi-Fi names. Users who connected could be redirected to fake login pages that collected email or social-media credentials.
- Prevention: users should verify the official network name, disable automatic connection to unknown networks, use a VPN where appropriate, and avoid entering sensitive credentials on suspicious Wi-Fi portals.
- Article: “Personal data stolen from unsuspecting airport visitors and plane passengers in ‘evil twin’ attacks, man charged” — July 1, 2024.



Prevention: verify the network, disable auto-connect, use VPN/HTTPS, avoid entering sensitive credentials on u